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Chapter 1

Theoretical Foundation

1.1 Basic Physical Quantities

1.1.1 Electrical Charge

1. Definition : **Charge** is an electrical property of the atomic particles of which matter consists, measured in coulombs(C).
2. Charge is **quantized** : the charge $-e$ of an electron is negative and $-e = -1.602 \times 10^{-19}C$, while a proton carries $+e$. The presence of equal numbers of protons and electrons leaves an atom neutrally charged. The only charges that occur in nature are integral multiples of the electronic charge e .
3. 1C is a large unit for charges : In 1C of charge, there are $1/(1.602 \times 10^{-19}C) \approx 6.24 \times 10^{18}$ electrons. Thus realistic or laboratory values of charges are on the order of pC, nC or μC .
4. **The law of conservation of charge** : charge can neither be created nor destroyed, only transferred. Thus the algebraic sum of the electric charges in a system does not change.

1.1.2 Electrical current

1. Electric current is the time rate of change of charge, measured in amperes(A).

$$i \equiv \frac{dq}{dt}$$

2. Unit of electric current : 1 A (ampere) = 1 C/s (Coulomb/second)
3. The charge transferred between time t_0 and t is

$$Q \equiv \int_{t_0}^t i \, dt$$

4. A direct current (DC) is a current that remains constant with time(i.e., $\frac{di}{dt} = 0$, while a alternating current (AC) is a current that varies sinusoidally with time.

1.1.3 Electrical Voltage

1. Definition : **Voltage** (or potential difference or electromotive force, emf) is the **energy** required to move a **unit charge** from a reference point (‘) to another point (+), measured in volts(V).
2. **Polarity** of Voltage : the voltage V_{ab} (the potential at point a with respect to point b) in an electric circuit is the energy (or work) needed to move a unit charge from a to b , as defined by

$$V_{ab} \equiv \frac{dW_{ab}}{dq}$$

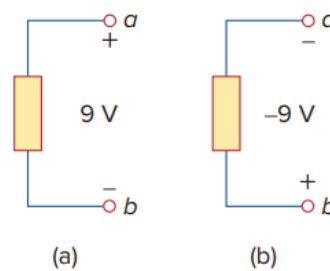


Figure 1.7

Two equivalent representations of the same voltage v_{ab} : (a) Point a is 9 V above point b ; (b) point b is -9 V above point a .

For example, in Fig. 1.7, we have two representations of the same voltage.

- In Fig. 1.7(a), point a is +9V above point b .
 - In Fig. 1.7(b), point b is -9V above point a .
 - There is a 9-V voltage drop from a to b or equivalently a 9 V voltage rise from b to a . In other words, a voltage drop from a to b is equivalent to a voltage rise from b to a .
3. Like electric current, a constant voltage is called a dc voltage and is represented by V , whereas a sinusoidally time-varying voltage is called an ac voltage and is represented by v .

1.1.4 Remarks

1. Current and voltage are the **two basic variables** in electric circuits.
2. The common term **signal** is used for an electric quantity such as a current or a voltage (or even electromagnetic wave) when it is used for **conveying information**. Engineers prefer to call such variables signals rather than mathematical functions of time because of their importance in communications and other disciplines.

1.1.5 Power and Energy

1. Motivation : Although electrical voltage and current are basic variables of an electric circuit, they are not sufficient by themselves. For instance, electrical bill that we must pay monthly is proportional to the amount of electrical energy we've used during the month.
2. Definition of **Power** is the time rate of expending or absorbing energy, measured in watts (W).

$$P \equiv \frac{dW}{dt} = \frac{dW}{dq} \cdot \frac{dq}{dt} = vi$$

Thus the power p is a time-varying quantity and is called the **instantaneous power**. In other words, the power absorbed ($p > 0$) or supplied ($p < 0$) by an element is the product of the voltage across the element and the current through it.

3. **Polarity** of the power : Passive sign convention is satisfied when the current enters through the **positive terminal** of an element and $p = +vi$. If the current enters through the negative terminal, $p = -vi$.

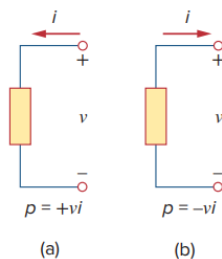


Figure 1.8

Reference polarities for power using the passive sign convention: (a) absorbing power, (b) supplying power.

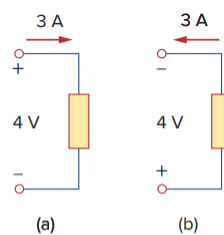


Figure 1.9

Two cases of an element with an absorbing power of 12 W: (a) $p = 4 \times 3 = 12$ W, (b) $p = 4 \times 3 = 12$ W.

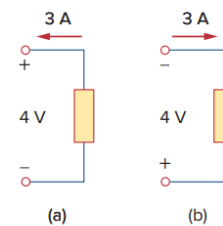


Figure 1.10

Two cases of an element with a supplying power of 12 W: (a) $p = -4 \times 3 = -12$ W, (b) $p = -4 \times 3 = -12$ W.

4. **The law of conservation of energy** : the algebraic sum of power in a circuit, at any instant of time, must be zero:

$$\sum p = 0$$

In other words, the total power supplied to the circuit must balance the total power absorbed.

5. Definition of **Energy** is the capacity to do work, measured in joules (J).

$$w = \int_{t_0}^t p \, dt = \int_{t_0}^t vi \, dt$$

6. The electric power utility companies measure energy in watt-hours (Wh), where $1Wh = 3600 J$

1.2 Basic circuit elements

1.2.1 Definitions

1. Elements : basic building blocks of a circuit.
 - (a) **Passive** elements cannot generate but consume energy. e.g., resistors, inductors, capacitors.
 - (b) **Active** elements generate or supply energy. e.g., generators, batteries, and operational amplifiers.
2. Circuit : interconnection of the elements
3. **Circuit analysis** : the process of determining voltages across (or the currents through) the elements of the circuit.

1.2.2 Most important active elements : Ideal voltage or current sources

There are two types of ideal voltage or current sources : independent and dependent sources :

(1) Ideal independent sources

1. Definition : An **ideal independent source** is an active element that provides a specified voltage or current that is completely independent of other circuit elements.
2. In other words, an ideal independent voltage source delivers to the circuit whatever current is necessary to maintain its terminal voltage. Physical sources such as **batteries and generators** may be regarded as approximations to ideal voltage sources. Likewise, the ideal independent current source delivers to the circuit whatever voltage is necessary to maintain the designated current.
3. The following figures depict the symbols for ideal independent voltage and current sources, respectively.

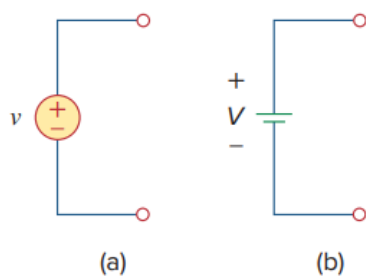


Figure 1.11
Symbols for independent voltage sources:
(a) used for constant or time-varying voltage, (b) used for constant voltage (dc).

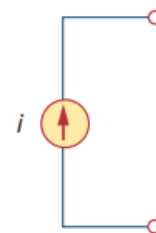


Figure 1.12
Symbol for independent current source.

(2) Ideal dependent sources

1. Definition : An **ideal dependent (or controlled) source** is an active element in which the source quantity is controlled by another voltage or current.
2. Dependent sources are usually designated by diamond-shaped symbols, as shown in the below figure:

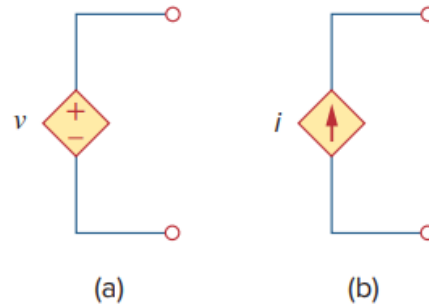


Figure 1.13
Symbols for: (a) dependent voltage source, (b) dependent current source.

3. Dependent sources are useful in modeling elements such as **transistors, operational amplifiers, and integrated circuits**.
4. Since the control of the dependent source is achieved by a voltage or current of some other element in the circuit, and the source can be voltage or current, it follows that there are four possible types of dependent sources, namely:
 - (a) A voltage-controlled voltage source (VCVS).
 - (b) A current-controlled voltage source (CCVS).
 - (c) A voltage-controlled current source (VCCS).
 - (d) A current-controlled current source (CCCS).

(3) Remarks : Energy considerations on Ideal sources

- (a) an ideal voltage source (dependent or independent) will produce any current required to ensure that the terminal voltage is as stated, whereas an ideal current source will produce the necessary voltage to ensure the stated current flow. Thus, an ideal source could in theory supply an **infinite amount of energy**.
- (b) Not only do sources supply power to a circuit, they can absorb power from a circuit too.
- (c) For a voltage source, we know the voltage but not the current supplied or drawn by it. By the same token, we know the current supplied by a current source but not the voltage across it.

1.2.3 Review exercises : Energy conservation law

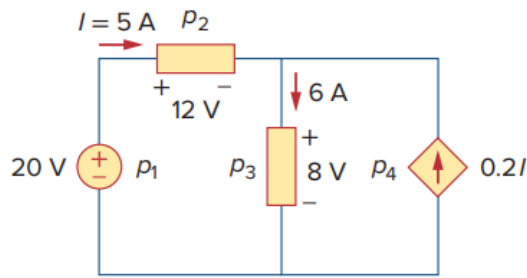


Figure 1.15
For Example 1.7.

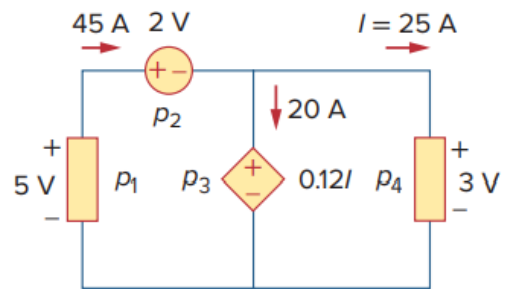


Figure 1.16
For Practice Prob. 1.7.

1. Calculate the power supplied or absorbed by each element in Fig. 1.15 and check whether this circuit satisfies energy conservation law.
2. Calculate the power supplied or absorbed by each element in Fig. 1.16 and check whether this circuit satisfies energy conservation law.

Solution. By passive sign convention, supplied powers are designated as negative whereas absorbed powers as positive.

1. **Ans.** $p_1 + p_2 + p_3 + p_4 = -100 \text{ W} + 60 \text{ W} + 48 \text{ W} - 8 \text{ W} = 0$. (energy conserved)
2. **Ans.** $p_1 + p_2 + p_3 + p_4 = -225 \text{ W} + 90 \text{ W} + 60 \text{ W} + 75 \text{ W} = 0$. (energy conserved)

1.3 Basic Laws

1.3.1 Resistance

1. **Resistor** : a circuit element used to model the current-resisting behavior of a material. The **resistance** of any material with a uniform cross-sectional area A depends on A and its length ℓ , as shown in Fig. 2.1(a)

$$R = \rho \frac{\ell}{A}$$

where ρ is the resistivity of the material.

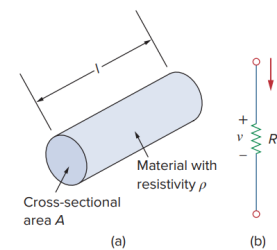


Figure 2.1
(a) Resistor, (b) Circuit symbol for resistance.

TABLE 2.1

Resistivities of common materials.

Material	Resistivity ($\Omega \cdot \text{m}$)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^2	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

2. **Ohm's law** : the voltage v across a resistor is directly proportional to the current i flowing through the resistor. That is,

$$v \propto i$$

The constant of proportionality is the resistance R and thus

$$v = iR$$

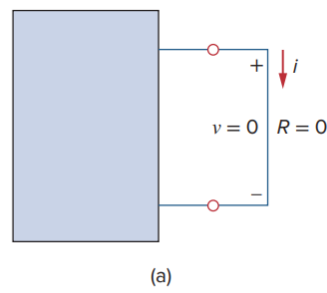
3. Two extreme cases of Ohm's law : Short and Open circuit

- (a) **Short** circuit : a circuit element with resistance approaching zero.

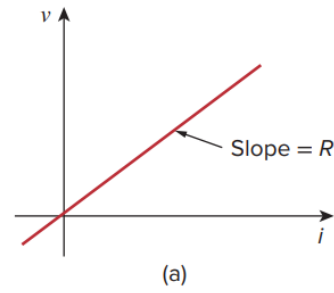
$$v = iR = i \times 0 = 0$$

Voltage is zero but the current could be anything. In practice, a short circuit is usually a connecting wire assumed to be a perfect conductor.

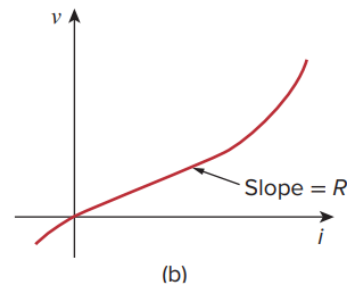
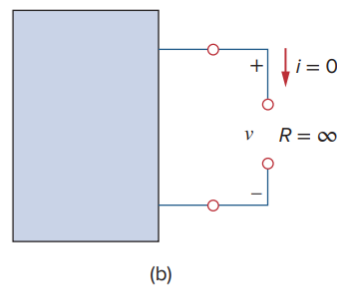
- (b) **Open** circuit : a circuit element with resistance approaching infinity. For open circuit, the current is zero though the voltage could be anything.

**Figure 2.2**

(a) Short circuit ($R = 0$), (b) Open circuit ($R = \infty$).

**Figure 2.7**

The i - v characteristic of: (a) a linear resistor, (b) a nonlinear resistor.



4. Resistors can also be classified as being linear or nonlinear depending on whether they obey Ohm's law or not (In Figure 2.7, $i - v$ characteristics of a linear and a nonlinear resistor are displayed, respectively.)
5. **Conductance** is defined as the reciprocal of the resistance and is a measure of how well an element will conduct electric current. The unit of conductance is Siemens ($1 \text{ S} = 1 \text{ A/V}$)

$$G \equiv \frac{1}{R} = \frac{i}{v}$$

Then, Ohm's law can be rewritten using conductance as $i = Gv$

6. The power dissipated by the resistor is

$$p = vi = i^2 R = \frac{v^2}{R}$$

Equivalently,

$$p = vi = v^2 G = \frac{i^2}{G}$$

Note that

- (a) The power dissipated in a resistor is a **nonlinear** function of either current or voltage.
- (b) Since R and G are positive quantities, the power dissipated in a resistor is always positive. Thus, a resistor **always absorbs** power from the circuit. This confirms the idea that a resistor is a **passive** element, incapable of generating energy.

1.3.2 Circuit as a Network Topology : Branch, nodes, and loops

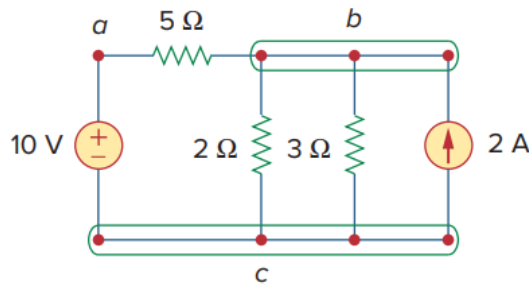


Figure 2.10
Nodes, branches, and loops.

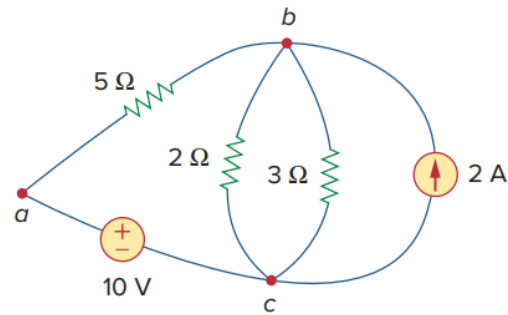


Figure 2.11
The three-node circuit of Fig. 2.10 is redrawn.

1. A **branch** represents a single element such as a voltage source or a resistor. In other words, a branch represents any two-terminal element.
2. A **node** is the point of connection between two or more branches. If a short circuit (a connecting wire) connects two nodes, the two nodes constitute a single node. (In Figure 2.10 and 2.11, three nodes a, b, c are depicted.)
3. A **loop** is any closed path in a circuit. A loop is a closed path formed by starting at a node, passing through a set of nodes, and returning to the starting node without passing through any node more than once.
 - (a) A loop is said to be **independent** if it contains at least one branch which is not a part of any other independent loop. Independent loops or paths result in **independent sets of equations**.
 - (b) It is possible to form an independent set of loops where one of the loops does not contain such a branch. In Fig. 2.11, $abca$ with the $2\ \Omega$ resistor is independent. A second loop bcb with the $3\ \Omega$ resistor and the current source is independent. The third loop bcb could be the one with the $2\ \Omega$ resistor in parallel with the $3\ \Omega$ resistor. This does form an independent set of loops.
4. **The Fundamental Theorem of Network Topology** : A network with b branches, n nodes, and l loops satisfies

$$b = n + l - 1$$
5. The simplest network topology of practical use : series and parallel connection
 - (a) Two or more elements are in **series** if they exclusively **share a single node** and consequently carry the same current.
 - (b) Two or more elements are in **parallel** if they are **connected to the same two nodes** and consequently have the same voltage across them.

1.3.3 Review Exercises : Network Topology

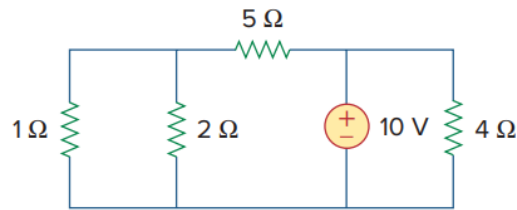


Figure 2.14
For Practice Prob. 2.4.

1. Check the fundamental theorem of network topology in the above circuit.
2. Identify the elements that are in series or parallel.

Solution.

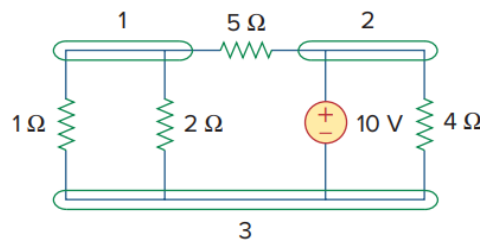


Figure 2.15
Answer for Practice Prob. 2.4.

1. $b = 5, n = 3, \ell = 3, b = n + \ell - 1$ satisfied.
2. 1옴과 2옴이 병렬, 10V 전압원과 4옴 저항 병렬.

1.3.4 Kirchhoff's Current law(KCL)

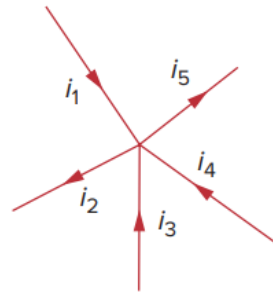


Figure 2.16
Currents at a node illustrating KCL.

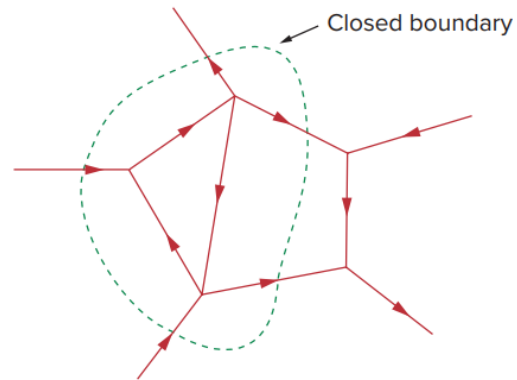


Figure 2.17
Applying KCL to a closed boundary.

1. Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a **node** (or a **closed boundary**) is zero.

$$\sum_{n=1}^N i_n = 0$$

where N is the number of branches connected to the node.

2. **Proof.** Consider a set of currents $i_k(t)$ ($k \in N$) flowing into a node. The algebraic sum of the node is

$$i_T(t) = i_1(t) + i_2(t) + i_3(t) + \dots$$

Now integrate both sides of the above,

$$q_T(t) = q_1(t) + q_2(t) + q_3(t) + \dots$$

where $q_k(t) = \int i_k(t)dt$ and $q_T(t) = \int i_T(t)dt$. **The law of conservation of charge** requires that the algebraic sum of electric charges at the node must not change; that is, the node stores no net charge. This implies $q_T(t) = 0$, which in turn concludes that $i_T(t) = 0$, as desired.

3. Alternatively, KCL states that the sum of the currents entering a node is equal to the sum of the currents leaving the node. More generally, as seen in Figure 2.17, KCL also applies to a closed boundary. This may be regarded as a generalized case, because a node may be regarded as a closed surface shrunk to a point. The total current entering the closed surface is equal to the total current leaving the surface.
4. A circuit cannot contain two different currents, I_1 and I_2 , in series, unless $I_1 = I_2$; otherwise KCL will be violated.

1.3.5 Kirchhoff's Voltage Law(KVL)

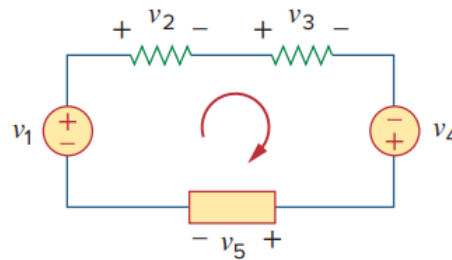


Figure 2.19

A single-loop circuit illustrating KVL.

1. Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

$$\sum_{n=1}^N v_n = 0$$

Here, N is the number of branches in the loop and v_n is the voltage across i^{th} branch.

2. For example, in Figure 2.19, we apply KVL in the counterclockwise direction starting from the voltage source v_1 . If we encounter (+) terminal, the voltage polarity of the branch is set positive. In this way, voltage drop is designated as positive whereas voltage rise as negative:

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

Alternatively,

$$v_2 + v_3 + v_5 = v_1 + v_4$$

Hence, KVL can also be casted as "Sum of voltage drops = Sum of voltage rises"

3. **Remark.** Although Sadiku tells us that KVL is a consequence of the law of energy conservation, this point needs some clarification.

1.3.6 Review Exercises: Applications of Ohm's law and Kirchhoff's law

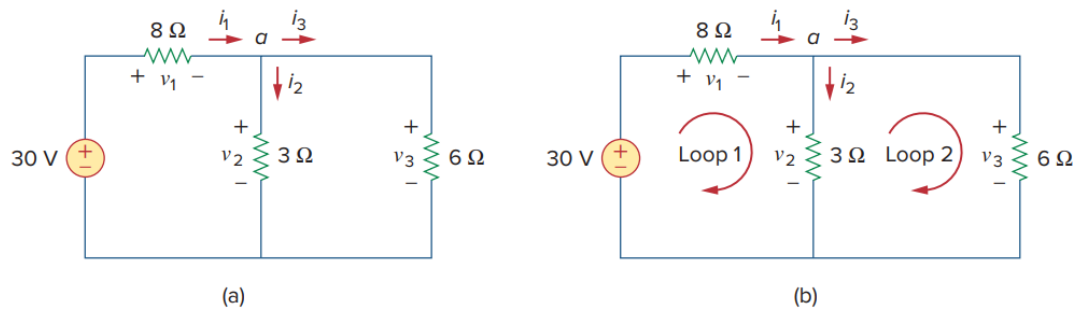


Figure 2.27
For Example 2.8.

Find currents and voltages in the circuit shown in Fig. 2.27

Solution.

1. Ohm's law : $v_1 = 8i_1$, $v_2 = 3i_2$, $v_3 = 6i_3$
2. KCL : $i_1 = i_2 + i_3$
3. KVL : $-30 + v_1 + v_2 = 0$ (loop 1), $-v_2 + v_3 = 0$ (loop 2).
4. **Ans.** $i_1 = 3A$, $i_2 = 2A$, $i_3 = 1A$, $v_1 = 24V$, $v_2 = v_3 = 6V$

1.3.7 Reducing a resistive network to a single equivalent resistance

A. Series resistance as a voltage divider

1. The equivalent resistance of any number of resistors connected in series is the sum of the individual resistances.

$$R_{eq} = R_1 + R_2 + \cdots + R_N$$

2. **Principle of Voltage division** : if a voltage divider with N resistors ($R_1, R_2, \cdots, R_n$) in series with the source voltage v , the voltage across n th resistor, v_n is

$$v_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} v$$

B. Parallel resistance as a current divider

1. When two resistors R_1, R_2 are connected in parallel and therefore have the same voltage across them, the equivalent resistance is equal to the product of their resistances divided by their sum.

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

2. When N resistors $R_1, R_2, \cdots, R_N$ are connected in parallel, the equivalent resistance is

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

3. Note that the R_{eq} is always **smaller than the resistance of the smallest resistor** in the parallel combination.
4. In a special case where $R_1 = R_2 = \cdots = R_N = R$, $R_{eq} = \frac{R}{N}$.
5. The conductance is more convenient when dealing with resistors in parallel. The equivalent conductance of resistors connected in parallel is the sum of their individual conductances.

$$G_{eq} = G_1 + G_2 + \cdots + G_N$$

where $G_k = 1/R_k (k = 1, 2, \cdots, N)$.

6. **Principle of current division** For two resistors in parallel, the equivalent resistor has the same voltage v ,

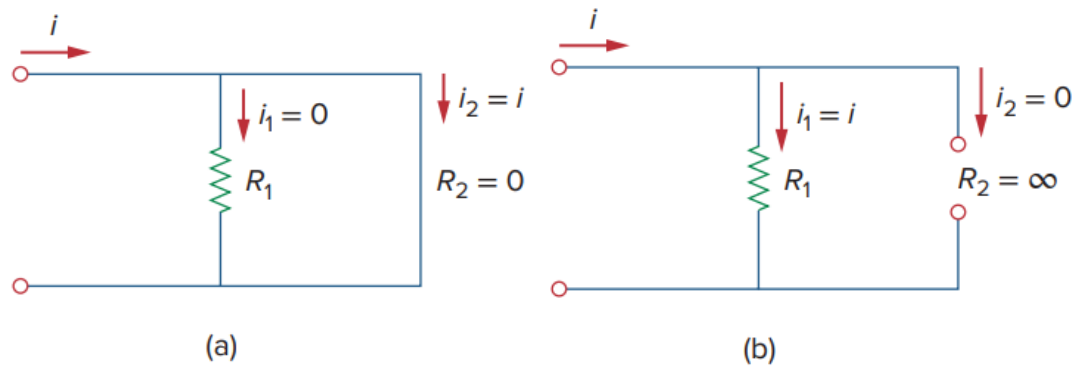
$$v = i R_{eq} = i \frac{R_1 R_2}{R_1 + R_2}$$

Since $v = i_1 R_1 = i_2 R_2$, the currents in each resistor is

$$i_1 = \frac{R_2}{R_1 + R_2} i, \quad i_2 = \frac{R_1}{R_1 + R_2} i$$

In other words, the total current i is divided by the resistors in inverse proportion to their resistances.

7. Let us investigate two extreme cases as depicted in the figure below:



- (a) In this case, R_2 is short-circuited. Entire current i bypasses R_1 and flows through the short circuit $R_2 = 0$, the path of least resistance. In other words, $R_{eq} = 0$ and the entire currents flows through the short circuit.
- (b) Now R_2 is open-circuited. The current still flows through the path of least resistance, R_1 . By taking the limit $R_2 \rightarrow \infty$, $R_{eq} \rightarrow R_1$ and $i = i_1$.
8. The principle of current division can be conveniently expressed using conductance,

$$i_n = \frac{G_n}{G_1 + G_2 + \cdots + G_N} i$$

Review Exercises : Equivalent resistance calculation

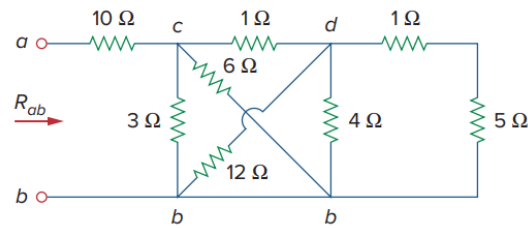
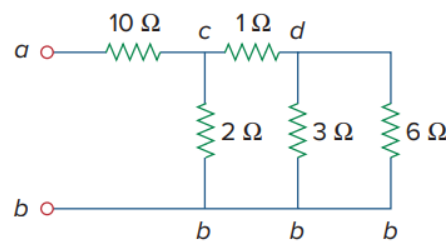


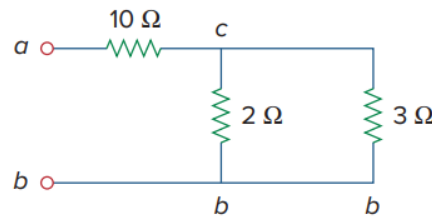
Figure 2.37
For Example 2.10.

Calculate the equivalent resistance in the Fig. 2.37.

Ans. $R_{eq} = 11.2 \, \Omega$ Here is the process reducing to simpler equivalent circuits



(a)



(b)

Figure 2.38
Equivalent circuits for Example 2.10.

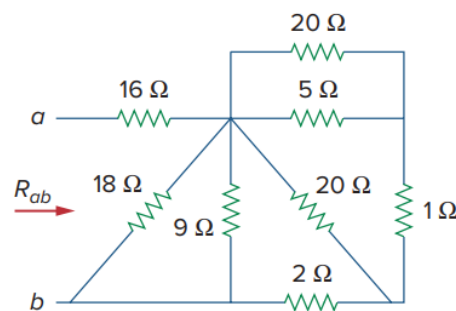


Figure 2.39
For Practice Prob. 2.10.

Ans. $R_{eq} = 19 \, \Omega$